

Kazakh British Technical University

Databases

Laboratory Work №3

DML

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Task 1 Generate and insert 200 random rows in your airport database

The screenshot shows the DBeaver interface with the 'airport_lab2' database selected in the Object Explorer. The 'public' schema is expanded, showing various database objects. The 'airline' table is highlighted. The SQL editor displays the following query:

```
1 INSERT INTO Passengers (  
2     passenger_id, first_name, last_name, date_of_birth,  
3     gender, country_of_citizenship, country_of_residence,  
4     passport_number, created_at, updated_at  
5 )  
6 SELECT  
7     generate_series(100, 299),  
8     'Name_' || generate_series(100, 299),  
9     'Surname_' || generate_series(100, 299),  
10    '1980-01-01'::DATE + (random() * 10000)::INT,  
11    CASE WHEN random() > 0.5 THEN 'Male' ELSE 'Female' END,  
12    'Kazakhstan',  
13    'KZ' || (100000 + generate_series(100, 299))::VARCHAR,  
14    CURRENT_TIMESTAMP,  
15    CURRENT_TIMESTAMP;  
16
```

The Data Output pane shows the execution result:

```
INSERT 0 200  
  
Query returned successfully in 397 msec.
```

A status message at the bottom right indicates: "Query returned successfully in 397 msec." The status bar shows "Total rows: Query complete 00:00:00.397" and "Ln 16, Col 23".

Task 2 Add a new airline named "KazAir" based in "Kazakhstan" to the airline table

The screenshot shows the DBeaver interface with the 'airport_lab2' database selected in the Object Explorer. The 'public' schema is expanded, showing various database objects. The 'airline' table is highlighted. The SQL editor displays the following query:

```
1 INSERT INTO Airline (airline_id, airline_code, airline_name, airline_country, cre  
2 VALUES (10, 'KZA', 'KazAir', 'Kazakhstan', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

The Data Output pane shows the execution result:

```
INSERT 0 1  
  
Query returned successfully in 103 msec.
```

A status message at the bottom right indicates: "Query returned successfully in 103 msec." The status bar shows "Total rows: Query complete 00:00:00.103" and "Ln 2, Col 82".

Task 3 Update the airline country "KazAir" to "Turkey".

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure for 'airport_lab2', including tables like 'airline', 'airport', and 'baggage'. The main query editor contains the following SQL statement:

```
1 UPDATE Airline
2 SET airline_country = 'Turkey'
3 WHERE airline_name = 'KazAir';
```

The 'Data Output' tab at the bottom shows the execution results:

```
UPDATE 1
Query returned successfully in 773 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 773 msec." The status bar also shows "Total rows: Query complete 00:00:00.773" and "LF Ln 3, Col 31".

Task 4 Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland"

The screenshot shows the PostgreSQL IDE interface. The main query editor contains the following SQL statement:

```
1 INSERT INTO Airline (airline_id, airline_code, airline_name, airline_country, cre
2 VALUES
3 (11, 'AE', 'AirEasy', 'France', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
4 (12, 'FH', 'FlyHigh', 'Brazil', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP),
5 (13, 'FF', 'FlyFly', 'Poland', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP);
```

The 'Data Output' tab at the bottom shows the execution results:

```
INSERT 0 3
Query returned successfully in 423 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 423 msec." The status bar also shows "Total rows: Query complete 00:00:00.423" and "LF Ln 5, Col 70".

Task 5 Delete all flights whose arrival in 2024 year.

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure for 'airport_lab2', including schemas, tables, and sequences. The main query editor contains the following SQL code:

```
1 DELETE FROM Flights
2 WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
```

The 'Messages' tab at the bottom shows the execution result: 'DELETE 0' and 'Query returned successfully in 274 msec.' A green notification box at the bottom right confirms: 'Query returned successfully in 274 msec.'

Task 6 Increase the price of all tickets in booking table for flights by 15%.

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure for 'airport_lab2'. The main query editor contains the following SQL code:

```
1 UPDATE Booking
2 SET ticket_price = ticket_price * 1.15;
```

The 'Messages' tab at the bottom shows the execution result: 'UPDATE 0' and 'Query returned successfully in 444 msec.' A green notification box at the bottom right confirms: 'Query returned successfully in 444 msec.'

Task 7 Delete all tickets where price is less than 10000.

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'airport_lab2' under 'PostgreSQL 18'. The 'Tables (15)' folder is expanded, showing tables like 'airline', 'airport', 'airport_3nf', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_3nf', 'booking_flight', 'booking_seat_3nf', and 'flight_3nf'. The main query window is titled 'airport_lab2/postgres@PostgreSQL 18' and contains the following SQL query:

```
1 DELETE FROM Booking
2 WHERE ticket_price < 10000;
```

The 'Data Output' tab is selected, showing the result of the query: 'DELETE 0'. Below this, a message states: 'Query returned successfully in 1 secs 376 msec.' A green status bar at the bottom right confirms: 'Query returned successfully in 1 secs 376 msec.' The status bar also shows 'Total rows: Query complete 00:00:01.376' and 'LF Ln 2, Col 28'.

Task 8 Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'airport_lab2' under 'PostgreSQL 18'. The 'Tables (15)' folder is expanded, showing tables like 'airline', 'airport', 'airport_3nf', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_3nf', 'booking_flight', 'booking_seat_3nf', and 'flight_3nf'. The main query window is titled 'airport_lab2/postgres@PostgreSQL 18' and contains the following SQL query:

```
1 UPDATE Airline
2 SET airline_code = 'UNK'
3 WHERE airline_code IS NULL;
```

The 'Data Output' tab is selected, showing the result of the query: 'UPDATE 0'. Below this, a message states: 'Query returned successfully in 2 secs 525 msec.' A green status bar at the bottom right confirms: 'Query returned successfully in 2 secs 525 msec.' The status bar also shows 'Total rows: Query complete 00:00:02.525' and 'LF Ln 3, Col 28'.

Task 9 Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.

The screenshot shows the DBeaver IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'airport_lab2', including tables like 'airline', 'airport', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_3nf', 'booking_flight', 'booking_seat_3nf', and 'flight_3nf'. The main query editor on the right contains the following SQL code:

```
1 DELETE FROM Baggage_check
2 WHERE created_at < '2023-06-01' AND check_result = 'Not checked';
```

Below the query editor, the 'Messages' pane shows the execution result: 'DELETE 0' and 'Query returned successfully in 684 msec.' A green status bar at the bottom right confirms: '✓ Query returned successfully in 684 msec.'

Task 10 Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.

The screenshot shows the DBeaver IDE interface. The 'Object Explorer' pane on the left is the same as in Task 9. The main query editor on the right contains the following SQL code:

```
1 DELETE FROM Airport
2 WHERE state IS NULL AND city IN ('Mlawe', 'Kepuh');
```

Below the query editor, the 'Messages' pane shows the execution result: 'DELETE 0' and 'Query returned successfully in 556 msec.' A green status bar at the bottom right confirms: '✓ Query returned successfully in 556 msec.'

Task 11 Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.

The screenshot shows the DBeaver IDE interface with a PostgreSQL database named 'airport_lab2'. The 'Query' tab is active, displaying a SQL script that inserts data into several tables and then inserts a new baggage check record. The 'Data Output' tab shows the result of the query, which is a single row with the following values:

baggage_check_id	created_at
999	2026-09-30 11:15:53.685224

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:01.989'.

Task 12 Convert all country names in the airline table to uppercase to ensure data consistency.

The screenshot shows the DBeaver IDE interface with the same PostgreSQL database. The 'Query' tab is active, displaying a SQL script that updates the 'airline' table to convert all country names to uppercase. The 'Messages' tab shows the result of the query, which is a single row with the following value:

UPDATE
5

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:03.387'. A green message box at the bottom right states: 'Query returned successfully in 3 secs 387 msec.'

Task 13 Update both the name and country for the airline with `airline_id = 5`. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating `update_at` to the current date.

The screenshot shows the DBeaver IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'PostgreSQL 18', including 'airport_lab2' and its tables. The main query editor on the right contains the following SQL statement:

```
1 UPDATE Airline
2 SET airline_name = 'Global Airways',
3     airline_country = 'United Kingdom',
4     updated_at = CURRENT_TIMESTAMP,
5 WHERE airline_id = 5;
```

Below the query editor, the 'Data Output' pane shows the execution results:

```
UPDATE 0
Query returned successfully in 552 msec.
```

A green status bar at the bottom right indicates: 'Query returned successfully in 552 msec.' The status bar also shows 'Total rows: Query complete 00:00:00.552' and 'LF Ln 5, Col 22'.

Task 14 Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'

The screenshot shows the DBeaver IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'PostgreSQL 18', including 'airport_lab2' and its tables. The main query editor on the right contains the following SQL statement:

```
1 UPDATE Airport
2 SET state = 'Capital District'
3 WHERE city IN ('Astana', 'London', 'Tokyo');
```

Below the query editor, the 'Data Output' pane shows the execution results:

```
UPDATE 0
Query returned successfully in 505 msec.
```

A green status bar at the bottom right indicates: 'Query returned successfully in 505 msec.' The status bar also shows 'Total rows: Query complete 00:00:00.505' and 'LF Ln 3, Col 45'.

Task 15 Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

Object Explorer

Servers (1)

PostgreSQL 18

Databases (3)

airport_lab2

Casts

Catalogs

Event Triggers

Extensions

Foreign Data Wrappers

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airport_3nf

baggage

baggage_check

boarding_pass

booking

booking_3nf

booking_flight

booking_seat_3nf

flight_3nf

phonebook_db/po... X

airport_lab2/postgres@PostgreSQL 18* X

airport_lab2/postgres@PostgreSQL 18

Query

Query History

AI Assistant

Scratch Pad X

1 UPDATE baggage_check

2 SET check_result = 'Checked'

3 WHERE EXTRACT(MONTH FROM created_at) = 3

4 AND EXTRACT(YEAR FROM created_at) = 2024

5 AND check_result = 'Not checked';

Data Output

Messages

Notifications

UPDATE 0

Query returned successfully in 487 msec.

Total rows: Query complete 00:00:00.487

LF Ln 5, Col 36

✓ Query returned successfully in 487 msec. ✕